

The Complete Theory of Consumption

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Abstract

Consumption is among the most fundamental activities of human civilization. Traditional economic theories explain consumption through utility maximization, income, wealth, expectations, and intertemporal optimization. Behavioural economics emphasizes psychological deviations from rationality. Finance studies the interaction between consumption, savings, debt, and wealth accumulation. Political economy examines how institutions and governments influence consumption. Warfare economics studies the relationship between consumption, savings, reserves, and national resilience.

This paper develops a unified framework called *The Complete Theory of Consumption*. The theory extends conventional consumption models by introducing two new state variables: *consumption credit* and *rationality credit*. Consumption credit allows individuals to consume beyond current income by borrowing from future resources. Rationality credit allows individuals to borrow decision-making capability from external institutions, educational systems, social structures, artificial intelligence systems, financial advisors, legal systems, and accumulated civilizational knowledge.

The resulting framework integrates economics, finance, psychology, philosophy, political science, international relations, geopolitics, welfare economics, warfare economics, statistics, computer science, artificial intelligence, and data science within a common mathematical structure.

The central thesis is that observed consumption emerges from the interaction of four fundamental forces:

$$\text{Consumption} = f(\text{Preferences, Rationality, Consumption Credit, Rationality Credit}).$$

The paper develops the mathematical foundations, derives theoretical results, constructs dynamic systems, introduces empirical estimation procedures, and proposes computational implementations suitable for large-scale simulation.

The paper ends with “The End”

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1 Introduction

Consumption occupies a central position in economics because virtually all economic activity ultimately serves consumption. Individuals work, save, invest, innovate, and engage in exchange largely to increase present or future consumption possibilities.

Classical theories of consumption emphasize rational utility maximization. Behavioural theories emphasize bounded rationality, psychological biases, and imperfect information. Modern economies additionally rely upon credit systems that permit consumption beyond current income.

Despite extensive literature, existing theories generally treat rationality as fixed, while credit is usually interpreted exclusively in financial terms.

This paper proposes a broader interpretation.

The theory introduces:

1. Rationality.
2. Propensity to consume.
3. Consumption credit.
4. Rationality credit.

Consumption credit represents the borrowing of financial resources.

Rationality credit represents the borrowing of cognitive resources.

The resulting framework extends traditional consumer theory into a unified theory of economic behaviour.

2 Historical Foundations

The theory builds upon several major intellectual traditions.

2.1 Classical Economics

Adam Smith emphasized self-interest and decentralized decision-making.

Consumption was viewed as the ultimate purpose of production.

2.2 Neoclassical Economics

Marshall, Jevons, Walras, and Pareto formalized utility maximization.

Consumers were assumed to solve optimization problems rationally.

2.3 Keynesian Economics

Keynes emphasized the marginal propensity to consume.

Consumption became linked to aggregate demand and macroeconomic fluctuations.

2.4 Life-Cycle and Permanent-Income Theories

Modigliani, Brumberg, and Friedman emphasized intertemporal consumption smoothing.

Consumers allocate resources across time rather than simply across goods.

2.5 Behavioural Economics

Simon, Kahneman, Tversky, and Thaler demonstrated that actual decision-making often departs from strict optimization.

2.6 Artificial Intelligence Era

Modern decision-making increasingly relies upon:

- Recommendation systems.
- Search engines.
- Financial algorithms.
- Artificial intelligence assistants.
- Automated advisory systems.

These systems create a new source of rationality external to the individual.

3 Fundamental Variables

The theory begins with four primitive state variables.

Definition 1 (Intrinsic Rationality). *Intrinsic rationality is*

$$r_0 \in [0, 1].$$

where:

- $r_0 = 0$ denotes complete irrationality.
- $r_0 = 1$ denotes perfect rationality.

Definition 2 (Propensity to Consume). *The propensity to consume is*

$$c \in [0, 1].$$

where:

- $c = 0$ denotes complete saving.
- $c = 1$ denotes complete consumption.

Definition 3 (Consumption Credit). *Consumption credit is*

$$\kappa \geq 0.$$

It measures consumption financed through borrowing rather than current income.

Definition 4 (Rationality Credit). *Rationality credit is*

$$\rho \geq 0.$$

It measures decision-making capability obtained from external systems.

4 The Complete Consumption State Space

The state space of the economy is

$$\Omega = [0, 1]^2 \times \mathbb{R}_+^2.$$

Each consumer occupies a point

$$(r_0, c, \kappa, \rho).$$

The dimensionality of the framework therefore expands beyond traditional consumer theory.

4.1 Interpretation

The first two coordinates describe internal behaviour.

The last two coordinates describe external support systems.

Thus the theory distinguishes between:

1. Internal cognitive resources.
2. External cognitive resources.
3. Internal financial resources.
4. External financial resources.

5 Behavioural Archetypes

The four-dimensional framework contains several important limiting cases.

r_0	κ	ρ	Interpretation
Low	Low	Low	Isolated subsistence consumer
High	Low	Low	Rational saver
Low	High	Low	Leveraged impulsive spender
High	High	Low	Sophisticated borrower
Low	Low	High	Institutionally supported consumer
High	High	High	Fully augmented optimizer

Table 1: Representative behavioural archetypes

6 Behavioural Geometry

The original behavioural framework occupied a unit square.

The complete framework occupies a four-dimensional manifold.

The first two dimensions form the behavioural plane.

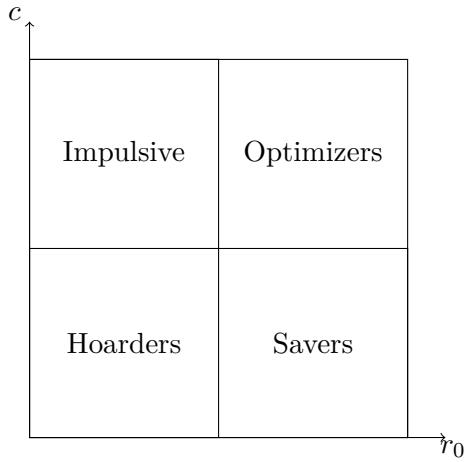


Figure 1: Behavioural state space in the (r_0, c) plane

The variables κ and ρ introduce two additional orthogonal dimensions.

7 Economic Interpretation of Credit

Traditional economics focuses primarily on financial credit.

This theory argues that all advanced societies operate simultaneously on:

1. Financial credit.
2. Informational credit.
3. Institutional credit.
4. Rationality credit.

Examples include:

- Schools.
- Universities.
- Courts.
- Accounting standards.
- Financial advisors.
- AI assistants.
- Search engines.
- Scientific institutions.

These systems permit individuals to make decisions superior to those implied by their intrinsic rationality alone.

8 A Civilizational Interpretation

Modern civilization can be viewed as a giant mechanism for the production of rationality credit.

Scientific institutions produce knowledge.

Educational institutions distribute knowledge.

Legal institutions codify knowledge.

Technological systems operationalize knowledge.

Individuals then consume this accumulated stock of rationality.

The resulting decision quality exceeds what any isolated individual could achieve independently.

This observation motivates the mathematical development that follows.

9 Preview of the Theory

The subsequent sections will derive:

1. Utility-theoretic foundations.
2. Optimal consumption rules.
3. Consumption-credit dynamics.
4. Rationality-credit dynamics.
5. Rationality debt.
6. Wealth accumulation systems.
7. Welfare theorems.
8. Political economy implications.
9. Geopolitical consequences.
10. Warfare-economic applications.
11. Statistical estimation procedures.
12. Artificial intelligence implementations.
13. Simulation algorithms.
14. General equilibrium extensions.

The central objective is to develop a mathematically complete theory capable of explaining consumption at the individual, institutional, national, and civilizational levels.

10 Utility Theory

The Complete Theory of Consumption begins with a utility-maximizing agent operating under both financial and cognitive constraints.

Let

$$Y_t$$

denote income,

$$C_t$$

denote consumption,
and

$$S_t$$

denote savings.

The conventional budget constraint is

$$Y_t = C_t + S_t.$$

Introducing consumption credit modifies the constraint to

$$Y_t + D_t = C_t + S_t,$$

where

$$D_t$$

is net borrowing.

Define consumption credit as

$$\kappa_t = \frac{D_t}{Y_t}.$$

Hence

$$C_t = c_t(1 + \kappa_t)Y_t.$$

Consumption is no longer restricted by current income.

10.1 Utility Function

Assume utility depends upon both consumption and saving.

Let

$$U(C_t, S_t) = \alpha \ln(C_t + \varepsilon) + (1 - \alpha) \ln(S_t + \varepsilon),$$

where

$$0 \leq \alpha \leq 1,$$

and

$$\varepsilon > 0.$$

Substituting

$$S_t = (1 + \kappa_t)Y_t - C_t,$$

gives

$$U(C_t) = \alpha \ln(C_t + \varepsilon) + (1 - \alpha) \ln\left((1 + \kappa_t)Y_t - C_t + \varepsilon\right).$$

11 Optimal Consumption

Differentiating,

$$\frac{\partial U}{\partial C_t} = \frac{\alpha}{C_t + \varepsilon} + \frac{1 - \alpha}{(1 + \kappa_t)Y_t - C_t + \varepsilon}.$$

Setting

$$\frac{\partial U}{\partial C_t} = 0,$$

yields

$$\alpha \left((1 + \kappa_t)Y_t - C_t + \varepsilon \right) = (1 - \alpha)(C_t + \varepsilon).$$

Neglecting small ε gives

$$\alpha(1 + \kappa_t)Y_t = C_t.$$

Thus:

Theorem 1 (Optimal Consumption Rule). *The optimal consumption level is*

$$C_t^* = \alpha(1 + \kappa_t)Y_t.$$

Proof. The first-order condition yields

$$\alpha((1 + \kappa_t)Y_t - C_t) = (1 - \alpha)C_t.$$

Collecting terms,

$$\alpha(1 + \kappa_t)Y_t = C_t.$$

Hence

$$C_t^* = \alpha(1 + \kappa_t)Y_t.$$

□

Therefore the optimal propensity to consume becomes

$$c_t^* = \alpha.$$

Credit expands the consumption frontier without altering the optimal fraction.

12 Consumption Credit Theory

Consumption credit allows individuals to transfer purchasing power from the future to the present.

Definition 5. *Consumption credit is*

$$\kappa_t = \frac{D_t}{Y_t}.$$

The effective budget set becomes

$$B_t = (1 + \kappa_t)Y_t.$$

12.1 Interpretation

κ	Interpretation
0	No borrowing
0.25	Borrowing 25% of income
0.50	Borrowing 50% of income
1.00	Borrowing income equal to annual income
> 1	Highly leveraged consumption

Table 2: Interpretation of consumption credit

12.2 Consumption Frontier

Maximum feasible consumption becomes

$$C_{\max} = (1 + \kappa)Y.$$

Therefore

$$\frac{\partial C_{\max}}{\partial \kappa} = Y.$$

Credit expands consumption possibilities linearly.

13 Rationality Credit Theory

The central innovation of this paper is the concept of rationality credit.

Definition 6. *Rationality credit is*

$$\rho \geq 0.$$

It represents cognitive capability borrowed from external systems.

Examples include:

- Education.
- Financial advisors.
- Legal institutions.
- Scientific knowledge.
- Artificial intelligence.
- Cultural norms.
- Social networks.
- Family guidance.

13.1 Effective Rationality

Let intrinsic rationality be

$$r_0.$$

Define effective rationality as

$$r = r_0 + \rho.$$

Since rationality cannot exceed perfection,

$$r = \min(1, r_0 + \rho).$$

14 Sources of Rationality Credit

Decompose rationality credit into components.

$$\rho = \rho_E + \rho_A + \rho_I + \rho_S + \rho_T,$$

where

$$\begin{aligned}\rho_E &= \text{Education Credit,} \\ \rho_A &= \text{Advisory Credit,} \\ \rho_I &= \text{Institutional Credit,} \\ \rho_S &= \text{Social Credit,} \\ \rho_T &= \text{Technological Credit.}\end{aligned}$$

Component	Example
ρ_E	Schooling, universities
ρ_A	Accountants, advisors
ρ_I	Courts, regulations
ρ_S	Family and communities
ρ_T	AI and algorithms

Table 3: Components of rationality credit

15 Behavioural Consumption Model

The original behavioural model becomes

$$c_t = (r_0 + \rho)c_t^* + (1 - r_0 - \rho)\eta_t,$$

where

$$\eta_t \sim U(0, 1).$$

The model combines:

1. Rational optimization.
2. Rationality augmentation.
3. Behavioural randomness.

16 Expectation and Variance

Since

$$E[\eta_t] = \frac{1}{2},$$

the expected consumption rate becomes

$$E[c_t] = (r_0 + \rho)c_t^* + (1 - r_0 - \rho)\frac{1}{2}.$$

Theorem 2. *Expected consumption converges toward optimal consumption as rationality credit increases.*

Proof. Differentiating,

$$\frac{\partial E[c_t]}{\partial \rho} = c_t^* - \frac{1}{2}.$$

Whenever

$$c_t^* > \frac{1}{2},$$

additional rationality credit increases expected consumption.

Whenever

$$c_t^* < \frac{1}{2},$$

additional rationality credit decreases expected consumption.

In both cases behaviour moves toward the optimum.

□

16.1 Variance

Because

$$\text{Var}(\eta_t) = \frac{1}{12},$$

we obtain

$$\text{Var}(c_t) = \frac{(1 - r_0 - \rho)^2}{12}.$$

Theorem 3 (Volatility Reduction). *Rationality credit reduces behavioural volatility.*

Proof. Differentiating,

$$\frac{\partial \text{Var}(c_t)}{\partial \rho} = -\frac{1 - r_0 - \rho}{6}.$$

For

$$0 \leq r_0 + \rho < 1,$$

the derivative is negative.

Therefore

$$\text{Var}(c_t)$$

decreases as rationality credit rises.

□

17 Behavioural Efficiency

Define behavioural efficiency by

$$B_t = 1 - |c_t - c_t^*|.$$

Clearly,

$$0 \leq B_t \leq 1.$$

Substituting the behavioural model yields

$$B_t = 1 - (1 - r_0 - \rho)|\eta_t - c_t^*|.$$

Therefore

$$E[B_t] = 1 - (1 - r_0 - \rho)E|\eta_t - c_t^*|.$$

Theorem 4. *Behavioural efficiency increases monotonically with rationality credit.*

Proof. Differentiating,

$$\frac{\partial E[B_t]}{\partial \rho} = E|\eta_t - c_t^*|.$$

Since absolute values are nonnegative,

$$E|\eta_t - c_t^*| \geq 0.$$

Hence

$$\frac{\partial E[B_t]}{\partial \rho} \geq 0.$$

□

18 The Rationality-Credit Theorem

Theorem 5. *Suppose*

$$c_t = (r_0 + \rho)c_t^* + (1 - r_0 - \rho)\eta_t.$$

Then:

1.

$$\lim_{\rho \rightarrow 1-r_0} E[c_t] = c_t^*.$$

2.

$$\lim_{\rho \rightarrow 1-r_0} \text{Var}(c_t) = 0.$$

3.

$$\lim_{\rho \rightarrow 1-r_0} B_t = 1.$$

Proof. Substitute

$$\rho = 1 - r_0$$

into the behavioural equation.

Then

$$c_t = c_t^*.$$

Therefore

$$E[c_t] = c_t^*,$$

and

$$Var(c_t) = 0.$$

Finally,

$$B_t = 1 - |c_t^* - c_t^*| = 1.$$

□

19 The Consumption-Rationality Duality

Consumption credit relaxes financial constraints.

Rationality credit relaxes cognitive constraints.

The two forms of credit therefore occupy dual positions within the theory.

Financial Dimension	Cognitive Dimension
Consumption Credit κ	Rationality Credit ρ
Debt	Advice
Borrowed Money	Borrowed Intelligence
Banks	Institutions
Interest Payments	Maintenance Costs
Default Risk	Cognitive Dependence

This duality forms the conceptual foundation for the dynamic theory developed in the next section.

20 Rationality Debt

The introduction of rationality credit naturally implies the possibility of rationality debt.

Just as excessive financial borrowing may weaken future balance sheets, excessive dependence upon external decision-making systems may weaken intrinsic decision-making capability.

Definition 7 (Rationality Debt). *Rationality debt is denoted*

$$\delta_t \geq 0.$$

It represents deterioration in intrinsic reasoning capacity arising from persistent dependence upon external cognitive systems.

Examples include:

- Blind reliance on advisors.
- Excessive dependence on algorithms.
- Delegation of judgment.
- Cognitive atrophy.
- Information outsourcing.

20.1 Intrinsic Rationality Dynamics

Let intrinsic rationality evolve according to

$$r_{0,t+1} = r_{0,t} + \lambda L_t - \mu \delta_t,$$

where

$$\begin{aligned} L_t &= \text{learning effort,} \\ \lambda &> 0, \\ \mu &> 0. \end{aligned}$$

The first term represents rationality accumulation.

The second term represents rationality depreciation.

Theorem 6. *If*

$$\lambda L_t > \mu \delta_t,$$

then intrinsic rationality increases.

Proof. Observe

$$r_{0,t+1} - r_{0,t} = \lambda L_t - \mu \delta_t.$$

The expression is positive whenever

$$\lambda L_t > \mu \delta_t.$$

□

21 Rationality Capital

Define the stock of accumulated rationality.

Definition 8. *Individual rationality capital is*

$$R_t.$$

Its evolution is

$$R_{t+1} = (1 - \omega)R_t + \lambda L_t + \rho_t,$$

where

$$0 < \omega < 1$$

is the depreciation rate.

Rationality capital therefore behaves analogously to physical capital.

21.1 Interpretation

Rationality capital includes:

- Education.
- Knowledge.
- Scientific literacy.
- Financial literacy.
- Strategic reasoning.
- Decision-making skill.
- Computational capability.

The theory therefore treats cognition as an economic asset.

22 Consumption-Credit Dynamics

Financial credit evolves through time.

Let

$$\kappa_t = \frac{D_t}{Y_t}.$$

Assume

$$\kappa_{t+1} = (1 + i)\kappa_t + \gamma(c_t - c_t^*),$$

where

$$\begin{aligned} i &= \text{interest rate,} \\ \gamma &> 0. \end{aligned}$$

The first term captures debt accumulation.

The second term captures behavioural overspending.

Theorem 7. *Persistent consumption above the optimum causes debt accumulation.*

Proof. If

$$c_t > c_t^*,$$

then

$$\gamma(c_t - c_t^*) > 0.$$

Hence

$$\kappa_{t+1} > (1 + i)\kappa_t.$$

Debt grows faster than compound interest alone.

□

23 The Consumption-Debt Spiral

Repeated overspending generates a feedback mechanism.

Suppose

$$c_t > c_t^*$$

for all t .

Then

$$\kappa_t \rightarrow \infty.$$

The economy enters a debt spiral.

Definition 9 (Consumption-Debt Spiral). *A consumption-debt spiral occurs whenever*

$$\lim_{t \rightarrow \infty} \kappa_t = \infty.$$

Debt spirals are common features of:

- Household crises.
- Banking crises.
- Sovereign debt crises.
- Asset bubbles.
- Credit booms.

24 Wealth Accumulation

Wealth evolves according to

$$W_{t+1} = (1 + i_t)(W_t + Y_t - C_t).$$

Substituting

$$C_t = c_t(1 + \kappa_t)Y_t$$

gives

$$W_{t+1} = (1 + i_t)[W_t + Y_t - c_t(1 + \kappa_t)Y_t].$$

Proposition 1. *Wealth accumulation decreases with both consumption propensity and consumption credit.*

Proof. Differentiating,

$$\frac{\partial W_{t+1}}{\partial c_t} = -(1 + i_t)(1 + \kappa_t)Y_t < 0.$$

Similarly,

$$\frac{\partial W_{t+1}}{\partial \kappa_t} = -(1 + i_t)c_t Y_t < 0.$$

□

25 Finance and Asset Pricing

Consumption influences financial valuation.

Suppose utility is intertemporal:

$$U = E \left[\sum_{t=0}^{\infty} \beta^t u(C_t) \right].$$

The stochastic discount factor becomes

$$M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

Asset prices satisfy

$$P_t = E_t [M_{t+1} X_{t+1}].$$

Because consumption depends upon rationality and credit,

$$P_t = f(r_t, \rho_t, \kappa_t).$$

Thus behavioural credit conditions influence asset valuation.

26 Behavioural Finance

Financial markets aggregate heterogeneous consumers.

Define market rationality:

$$\bar{r} = \frac{1}{N} \sum_{i=1}^N (r_{0,i} + \rho_i).$$

Low market rationality may generate:

- Bubbles.
- Manias.
- Crashes.
- Panic selling.
- Excess volatility.

Theorem 8. *Market volatility decreases as average rationality increases.*

Proof. Individual variance is

$$\text{Var}(c_i) = \frac{(1 - r_i)^2}{12}.$$

Averaging across consumers implies lower aggregate variance for larger average rationality. □

27 Macroeconomic Aggregation

Consider a population of N consumers.

Average consumption is

$$\bar{C} = \frac{1}{N} \sum_{i=1}^N C_i.$$

Average rationality is

$$\bar{r} = \frac{1}{N} \sum_{i=1}^N (r_{0,i} + \rho_i).$$

Average consumption credit is

$$\bar{\kappa} = \frac{1}{N} \sum_{i=1}^N \kappa_i.$$

27.1 Aggregate Consumption Function

A linear approximation is

$$\bar{C} = \beta_0 + \beta_1 \bar{r} + \beta_2 \bar{\kappa} + \beta_3 \bar{Y}.$$

Proposition 2. *Aggregate consumption is increasing in both rationality and credit availability.*

Proof. Provided

$$\beta_1 > 0 \quad \text{and} \quad \beta_2 > 0,$$

we have

$$\frac{\partial \bar{C}}{\partial \bar{r}} = \beta_1 > 0,$$

and

$$\frac{\partial \bar{C}}{\partial \bar{\kappa}} = \beta_2 > 0.$$

□

28 National Rationality Capital

The theory extends naturally to nations.

Definition 10. *National rationality capital is*

$$R_N = \sum_{i=1}^N \rho_i.$$

This stock measures the rationality infrastructure of a society.
Examples include:

- Universities.
- Research institutes.

- Statistical agencies.
- Courts.
- Accounting systems.
- Educational institutions.
- Scientific communities.
- Artificial intelligence systems.

28.1 National Output

Suppose production is

$$Y = AK^\alpha L^\beta R_N^\theta.$$

Then

$$\frac{\partial Y}{\partial R_N} = \theta AK^\alpha L^\beta R_N^{\theta-1}.$$

Theorem 9. *National rationality capital raises productive capacity whenever*

$$\theta > 0.$$

Proof. Since

$$A > 0, \quad K > 0, \quad L > 0, \quad R_N > 0,$$

the derivative is positive whenever

$$\theta > 0.$$

□

29 The Rationality-Wealth Theorem

Theorem 10. *Suppose*

$$W_{t+1} = (1+i) \left[W_t + Y_t - c_t(1+\kappa_t)Y_t \right]$$

and

$$c_t = (r_0 + \rho)c_t^* + (1 - r_0 - \rho)\eta_t.$$

Then increasing rationality credit reduces expected behavioural wealth losses.

Proof. Expected consumption is

$$E[c_t] = (r_0 + \rho)c_t^* + (1 - r_0 - \rho)\frac{1}{2}.$$

As ρ increases, consumption converges toward the optimum.

Behavioural deviations shrink.

Therefore expected wealth losses arising from irrational decisions decline.

□

30 Four Macroeconomic Regimes

The theory identifies four fundamental economic regimes.

Regime	κ	ρ	Characterization
I	Low	High	Prudent society
II	High	High	Sophisticated leveraged society
III	Low	Low	Underdeveloped society
IV	High	Low	Fragile debt-driven society

Table 4: Macroeconomic consumption regimes

Regime IV is the most unstable because high borrowing combines with weak decision-making capability.

31 Rationality Debt

The introduction of rationality credit naturally implies the possibility of rationality debt.

Just as excessive financial borrowing may weaken future balance sheets, excessive dependence upon external decision-making systems may weaken intrinsic decision-making capability.

Definition 11 (Rationality Debt). *Rationality debt is denoted*

$$\delta_t \geq 0.$$

It represents deterioration in intrinsic reasoning capacity arising from persistent dependence upon external cognitive systems.

Examples include:

- Blind reliance on advisors.
- Excessive dependence on algorithms.
- Delegation of judgment.
- Cognitive atrophy.
- Information outsourcing.

31.1 Intrinsic Rationality Dynamics

Let intrinsic rationality evolve according to

$$r_{0,t+1} = r_{0,t} + \lambda L_t - \mu \delta_t,$$

where

$$\begin{aligned} L_t &= \text{learning effort,} \\ \lambda &> 0, \\ \mu &> 0. \end{aligned}$$

The first term represents rationality accumulation.

The second term represents rationality depreciation.

Theorem 11. *If*

$$\lambda L_t > \mu \delta_t,$$

then intrinsic rationality increases.

Proof. Observe

$$r_{0,t+1} - r_{0,t} = \lambda L_t - \mu \delta_t.$$

The expression is positive whenever

$$\lambda L_t > \mu \delta_t.$$

□

32 Rationality Capital

Define the stock of accumulated rationality.

Definition 12. *Individual rationality capital is*

$$R_t.$$

Its evolution is

$$R_{t+1} = (1 - \omega)R_t + \lambda L_t + \rho_t,$$

where

$$0 < \omega < 1$$

is the depreciation rate.

Rationality capital therefore behaves analogously to physical capital.

32.1 Interpretation

Rationality capital includes:

- Education.
- Knowledge.
- Scientific literacy.
- Financial literacy.
- Strategic reasoning.
- Decision-making skill.
- Computational capability.

The theory therefore treats cognition as an economic asset.

33 Consumption-Credit Dynamics

Financial credit evolves through time.

Let

$$\kappa_t = \frac{D_t}{Y_t}.$$

Assume

$$\kappa_{t+1} = (1 + i)\kappa_t + \gamma(c_t - c_t^*),$$

where

$$\begin{aligned} i &= \text{interest rate,} \\ \gamma &> 0. \end{aligned}$$

The first term captures debt accumulation.

The second term captures behavioural overspending.

Theorem 12. *Persistent consumption above the optimum causes debt accumulation.*

Proof. If

$$c_t > c_t^*,$$

then

$$\gamma(c_t - c_t^*) > 0.$$

Hence

$$\kappa_{t+1} > (1 + i)\kappa_t.$$

Debt grows faster than compound interest alone.

□

34 The Consumption-Debt Spiral

Repeated overspending generates a feedback mechanism.

Suppose

$$c_t > c_t^*$$

for all t .

Then

$$\kappa_t \rightarrow \infty.$$

The economy enters a debt spiral.

Definition 13 (Consumption-Debt Spiral). *A consumption-debt spiral occurs whenever*

$$\lim_{t \rightarrow \infty} \kappa_t = \infty.$$

Debt spirals are common features of:

- Household crises.
- Banking crises.
- Sovereign debt crises.
- Asset bubbles.
- Credit booms.

35 Wealth Accumulation

Wealth evolves according to

$$W_{t+1} = (1 + i_t)(W_t + Y_t - C_t).$$

Substituting

$$C_t = c_t(1 + \kappa_t)Y_t$$

gives

$$W_{t+1} = (1 + i_t)[W_t + Y_t - c_t(1 + \kappa_t)Y_t].$$

Proposition 3. *Wealth accumulation decreases with both consumption propensity and consumption credit.*

Proof. Differentiating,

$$\frac{\partial W_{t+1}}{\partial c_t} = -(1 + i_t)(1 + \kappa_t)Y_t < 0.$$

Similarly,

$$\frac{\partial W_{t+1}}{\partial \kappa_t} = -(1 + i_t)c_tY_t < 0.$$

□

36 Finance and Asset Pricing

Consumption influences financial valuation.

Suppose utility is intertemporal:

$$U = E \left[\sum_{t=0}^{\infty} \beta^t u(C_t) \right].$$

The stochastic discount factor becomes

$$M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

Asset prices satisfy

$$P_t = E_t [M_{t+1} X_{t+1}].$$

Because consumption depends upon rationality and credit,

$$P_t = f(r_t, \rho_t, \kappa_t).$$

Thus behavioural credit conditions influence asset valuation.

37 Behavioural Finance

Financial markets aggregate heterogeneous consumers.

Define market rationality:

$$\bar{r} = \frac{1}{N} \sum_{i=1}^N (r_{0,i} + \rho_i).$$

Low market rationality may generate:

- Bubbles.
- Manias.
- Crashes.
- Panic selling.
- Excess volatility.

Theorem 13. *Market volatility decreases as average rationality increases.*

Proof. Individual variance is

$$\text{Var}(c_i) = \frac{(1 - r_i)^2}{12}.$$

Averaging across consumers implies lower aggregate variance for larger average rationality. $\square$

38 Macroeconomic Aggregation

Consider a population of N consumers.

Average consumption is

$$\bar{C} = \frac{1}{N} \sum_{i=1}^N C_i.$$

Average rationality is

$$\bar{r} = \frac{1}{N} \sum_{i=1}^N (r_{0,i} + \rho_i).$$

Average consumption credit is

$$\bar{\kappa} = \frac{1}{N} \sum_{i=1}^N \kappa_i.$$

38.1 Aggregate Consumption Function

A linear approximation is

$$\bar{C} = \beta_0 + \beta_1 \bar{r} + \beta_2 \bar{\kappa} + \beta_3 \bar{Y}.$$

Proposition 4. *Aggregate consumption is increasing in both rationality and credit availability.*

Proof. Provided

$$\beta_1 > 0 \quad \text{and} \quad \beta_2 > 0,$$

we have

$$\frac{\partial \bar{C}}{\partial \bar{r}} = \beta_1 > 0,$$

and

$$\frac{\partial \bar{C}}{\partial \bar{\kappa}} = \beta_2 > 0.$$

□

39 National Rationality Capital

The theory extends naturally to nations.

Definition 14. *National rationality capital is*

$$R_N = \sum_{i=1}^N \rho_i.$$

This stock measures the rationality infrastructure of a society. Examples include:

- Universities.
- Research institutes.
- Statistical agencies.
- Courts.
- Accounting systems.
- Educational institutions.
- Scientific communities.
- Artificial intelligence systems.

39.1 National Output

Suppose production is

$$Y = AK^\alpha L^\beta R_N^\theta.$$

Then

$$\frac{\partial Y}{\partial R_N} = \theta AK^\alpha L^\beta R_N^{\theta-1}.$$

Theorem 14. *National rationality capital raises productive capacity whenever*

$$\theta > 0.$$

Proof. Since

$$A > 0, \quad K > 0, \quad L > 0, \quad R_N > 0,$$

the derivative is positive whenever

$$\theta > 0.$$

□

40 The Rationality-Wealth Theorem

Theorem 15. *Suppose*

$$W_{t+1} = (1 + i) \left[W_t + Y_t - c_t(1 + \kappa_t)Y_t \right]$$

and

$$c_t = (r_0 + \rho)c_t^* + (1 - r_0 - \rho)\eta_t.$$

Then increasing rationality credit reduces expected behavioural wealth losses.

Proof. Expected consumption is

$$E[c_t] = (r_0 + \rho)c_t^* + (1 - r_0 - \rho)\frac{1}{2}.$$

As ρ increases, consumption converges toward the optimum.

Behavioural deviations shrink.

Therefore expected wealth losses arising from irrational decisions decline.

□

41 Four Macroeconomic Regimes

The theory identifies four fundamental economic regimes.

Regime	κ	ρ	Characterization
I	Low	High	Prudent society
II	High	High	Sophisticated leveraged society
III	Low	Low	Underdeveloped society
IV	High	Low	Fragile debt-driven society

Table 5: Macroeconomic consumption regimes

Regime IV is the most unstable because high borrowing combines with weak decision-making capability.

42 Welfare Economics

Consumption is ultimately relevant because it contributes to human welfare.

Let social welfare be defined as

$$SW = \sum_{i=1}^N U_i.$$

Substituting the utility function,

$$SW = \sum_{i=1}^N [\alpha_i \ln(C_i + \varepsilon) + (1 - \alpha_i) \ln(S_i + \varepsilon)].$$

The Complete Theory of Consumption introduces a new welfare channel:

$$\rho_i.$$

Individuals may experience welfare gains through improved decision-making, even if income remains unchanged.

42.1 Behavioural Welfare Loss

Define welfare loss as

$$L_i = U(C_i^*) - U(C_i).$$

Since

$$C_i^*$$

maximizes utility,

$$L_i \geq 0.$$

Theorem 16. *Expected welfare loss decreases as rationality credit increases.*

Proof. The behavioural consumption equation is

$$c_i = (r_{0,i} + \rho_i)c_i^* + (1 - r_{0,i} - \rho_i)\eta_i.$$

As

$$\rho_i$$

increases,

$$c_i \rightarrow c_i^*.$$

Therefore

$$L_i \rightarrow 0.$$

□

42.2 Behavioural Welfare Frontier

Define

$$W_i = f(r_i, \kappa_i, \rho_i).$$

The welfare frontier represents all attainable combinations of:

- Consumption.
- Saving.

- Rationality.
- Credit.

The frontier expands whenever:

1. Income increases.
2. Rationality credit increases.
3. Productive credit increases.
4. Knowledge improves.

43 Public Policy

Governments influence consumption through:

1. Taxation.
2. Transfers.
3. Education.
4. Regulation.
5. Credit markets.
6. Monetary policy.

Within the present framework policy affects:

$$Y, \quad \kappa, \quad \rho.$$

A government therefore influences both financial and cognitive constraints.

43.1 Policy Efficiency

Define policy efficiency as

$$PE = \frac{\Delta SW}{G},$$

where

$$G$$

denotes government expenditure.

Policies that increase rationality credit may produce persistent welfare gains.

Examples include:

- Financial literacy programs.
- Educational reform.
- Statistical transparency.
- Scientific investment.
- Public information campaigns.

44 Political Economy

Political systems influence both credit allocation and rationality production.

Let

$$P_t$$

denote political power.

Suppose

$$\rho_t = g(P_t).$$

Different political institutions produce different quantities of rationality credit.

44.1 The Political Rationality Hypothesis

Political institutions perform two functions:

1. Resource allocation.
2. Rationality allocation.

Traditional political economy focuses primarily on the first.

The Complete Theory of Consumption argues that the second is equally important.

Definition 15. *State rationality production is*

$$SR_t.$$

Examples include:

- Public education.
- Scientific research.
- Statistical agencies.
- Legal systems.
- Regulatory institutions.

45 Democracy, Autocracy, and Rationality Credit

Suppose

$$\rho = \rho(P).$$

Different institutional arrangements may produce different levels of rationality credit.

The theory does not assume any specific political structure is universally optimal.

Instead it predicts that societies generating larger stocks of effective rationality capital possess advantages in:

- Economic planning.
- Innovation.
- Crisis management.
- Capital allocation.
- Long-run development.

46 The Political Consumption Equation

Aggregate consumption becomes

$$\bar{C} = f(\bar{Y}, \bar{r}, \bar{\kappa}, \bar{\rho}, P).$$

Political systems therefore influence consumption both directly and indirectly.

47 International Relations

Consumption is not merely an individual phenomenon.

Consumption patterns influence international power structures.

Define:

$$NX = X - M,$$

where

X = exports,

M = imports.

47.1 Consumption and Trade

Suppose imports depend upon aggregate consumption:

$$M = m(\bar{C}).$$

Then

$$\frac{\partial M}{\partial \bar{C}} > 0.$$

Higher consumption may increase import dependence.

Theorem 17. *Consumption-driven economies may experience declining trade balances.*

Proof. As consumption rises,

$$M$$

increases.

If exports remain unchanged,

$$NX = X - M$$

falls.

□

48 Geopolitics

Geopolitical power depends partly upon productive capacity and partly upon strategic autonomy.

Define geopolitical autonomy:

$$A_G.$$

Assume

$$A_G = h(S, K, R_N),$$

where

S = savings,

K = productive capital,

R_N = national rationality capital.

48.1 Strategic Dependence

Excessive consumption may reduce:

- Domestic savings.
- Strategic reserves.
- Industrial redundancy.
- Supply-chain resilience.

Consequently

$$A_G$$

may decline.

49 The Geopolitical Consumption Tradeoff

A nation faces a tradeoff between:

1. Current consumption.
2. Future strategic flexibility.

Let

$$G_t = \phi C_t + (1 - \phi) A_G.$$

The optimal nation maximizes both simultaneously.

50 National Resilience

Define national resilience as

$$R_s.$$

Assume

$$R_s = f(S, K, R_N, M_s),$$

where

$$M_s$$

denotes strategic material reserves.

Examples include:

- Food reserves.
- Energy reserves.
- Industrial inventories.
- Financial reserves.
- Military stockpiles.

51 Warfare Economics

The theory naturally extends into warfare economics.

Wars are periods during which:

1. Consumption falls.
2. Savings become strategic assets.
3. Reserves become critical.
4. Rationality becomes exceptionally valuable.

51.1 Strategic Consumption

Define wartime consumption:

$$C_W.$$

Typically

$$C_W < C_P,$$

where

$$C_P$$

denotes peacetime consumption.

Resources are redirected toward:

- Defense.

- Logistics.
- Production.
- Strategic reserves.

52 Military Readiness

Let

$$M_R$$

denote military readiness.

Assume

$$M_R = f(K, S, R_N).$$

Theorem 18. *Military readiness increases with national rationality capital.*

Proof. Suppose

$$\frac{\partial M_R}{\partial R_N} > 0.$$

Then increases in

$$R_N$$

improve planning, logistics, coordination, and technology utilization.

Therefore

$$M_R$$

rises.

□

53 The Strategic Rationality Theorem

Theorem 19. *Nations possessing larger stocks of rationality capital achieve greater strategic efficiency.*

Proof. Let strategic output be

$$Q_s = AK^\alpha L^\beta R_N^\theta.$$

Then

$$\frac{\partial Q_s}{\partial R_N} = \theta AK^\alpha L^\beta R_N^{\theta-1}.$$

For

$$\theta > 0,$$

the derivative is positive.

Hence larger rationality capital increases strategic output.

□

54 Civilizational Consumption

The framework may be interpreted at the scale of civilizations.

Let

$$\mathcal{C}$$

denote a civilization.

Its aggregate rationality capital is

$$R_{\mathcal{C}}.$$

Its aggregate consumption is

$$C_{\mathcal{C}}.$$

Its aggregate productive capacity is

$$Y_{\mathcal{C}}.$$

A civilization survives when:

$$Y_{\mathcal{C}} > C_{\mathcal{C}}.$$

for sufficiently long horizons.

54.1 The Civilizational Balance Condition

Define

$$B_{\mathcal{C}} = Y_{\mathcal{C}} - C_{\mathcal{C}}.$$

Then:

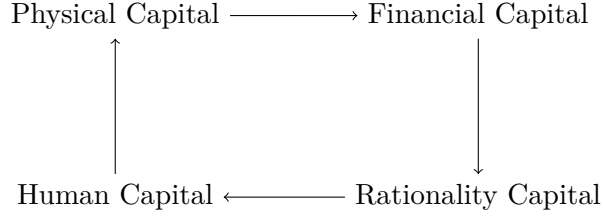
- $B_{\mathcal{C}} > 0$ implies accumulation.
- $B_{\mathcal{C}} = 0$ implies stagnation.
- $B_{\mathcal{C}} < 0$ implies decumulation.

55 The Four Capitals of Civilization

The Complete Theory of Consumption identifies four fundamental forms of capital:

1. Physical Capital.
2. Financial Capital.
3. Human Capital.
4. Rationality Capital.

Their interaction determines long-run prosperity.



The cycle illustrates the interdependence of productive systems.

56 Toward a General Theory of Civilization

Consumption is not merely expenditure.

It is the interface between:

- Human preferences.
- Cognitive systems.
- Financial systems.
- Political systems.
- Strategic systems.
- Civilizational systems.

The Complete Theory of Consumption therefore links microeconomic behavior to macroeconomic performance, geopolitical power, and long-run civilizational development.

57 Statistics and Econometric Foundations

The Complete Theory of Consumption is designed to be empirically testable.

Suppose data are observed for N consumers across T periods.

The observed dataset is

$$\mathcal{D} = c_{it}, r_{0,it}, \rho_{it}, \kappa_{it}, Y_{it}.$$

The fundamental behavioural equation is

$$c_{it} = (r_{0,it} + \rho_{it})c_{it}^* + (1 - r_{0,it} - \rho_{it})\eta_{it}.$$

The objective of empirical analysis is to estimate:

1. Intrinsic rationality.
2. Rationality credit.
3. Consumption credit.
4. Behavioural efficiency.
5. Welfare effects.

58 Linear Regression Framework

The simplest specification is

$$c_i = \beta_0 + \beta_1 r_{0,i} + \beta_2 \rho_i + \beta_3 \kappa_i + \varepsilon_i.$$

The parameter vector is

$$\beta = (\beta_0, \beta_1, \beta_2, \beta_3)'$$

Ordinary least squares yields

$$\hat{\beta} = (X'X)^{-1}X'Y.$$

58.1 Interpretation

Coefficient	Interpretation
β_1	Effect of intrinsic rationality
β_2	Effect of rationality credit
β_3	Effect of consumption credit

Table 6: Regression coefficients

59 Panel Data Models

Consumption evolves through time.

The panel specification becomes

$$c_{it} = \alpha_i + \beta_1 r_{0,it} + \beta_2 \rho_{it} + \beta_3 \kappa_{it} + u_{it}.$$

Fixed effects capture unobserved heterogeneity.

Random effects provide an alternative specification.

59.1 Dynamic Consumption Equation

A dynamic model is

$$c_{it} = \lambda c_{i,t-1} + \beta_1 r_{0,it} + \beta_2 \rho_{it} + \beta_3 \kappa_{it} + u_{it}.$$

This allows habit persistence.

60 Structural Estimation

The behavioural equation can be estimated directly.

Suppose

$$c_i = (r_{0,i} + \rho_i)c_i^* + (1 - r_{0,i} - \rho_i)\eta_i.$$

Observed consumption identifies:

$$r_{0,i} + \rho_i.$$

Additional instruments may separate intrinsic rationality from rationality credit.

Possible instruments include:

- Educational attainment.
- Institutional quality.
- Access to advisory services.
- AI utilization.
- Financial literacy.

61 Maximum Likelihood Estimation

Suppose

$$\eta_i \sim U(0, 1).$$

The likelihood becomes

$$L(\theta) = \prod_{i=1}^N f(c_i|\theta).$$

The log-likelihood is

$$\ell(\theta) = \sum_{i=1}^N \log f(c_i|\theta).$$

The estimator satisfies

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta).$$

62 Bayesian Estimation

Let

$$\theta = (r_0, \rho, \kappa).$$

Bayes' theorem yields

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)}.$$

Posterior distributions quantify uncertainty regarding:

- Rationality.
- Credit usage.
- Behavioural efficiency.
- Welfare losses.

63 Behavioural Efficiency Measurement

Recall

$$B_i = 1 - |c_i - c_i^*|.$$

The sample estimator is

$$\hat{B} = \frac{1}{N} \sum_{i=1}^N B_i.$$

Definition 16. *The Behavioural Efficiency Index is*

$$BEI = 100\hat{B}.$$

The index ranges from

$$0$$

to

$$100.$$

64 Consumption Volatility

Consumption volatility is

$$\sigma_C^2 = \text{Var}(C).$$

The model predicts

$$\sigma_C^2 = \frac{(1 - r_0 - \rho)^2}{12}.$$

Theorem 20. *Consumption volatility decreases with rationality credit.*

Proof. Differentiating,

$$\frac{\partial \sigma_C^2}{\partial \rho} = -\frac{1 - r_0 - \rho}{6}.$$

For

$$r_0 + \rho < 1,$$

the derivative is negative.

□

65 Data Science Framework

The Complete Theory of Consumption naturally generates large datasets.

Each observation may contain:

$$x_i = (r_{0,i}, \rho_i, \kappa_i, Y_i, W_i, Age_i, Edu_i).$$

The target variable is

$$c_i.$$

Data science methods can estimate nonlinear consumption functions.

66 Feature Engineering

Important engineered variables include:

$$\begin{aligned}RC_i &= r_{0,i} + \rho_i, \\CD_i &= \kappa_i Y_i, \\RD_i &= \delta_i, \\BE_i &= B_i.\end{aligned}$$

These summarize the cognitive and financial dimensions of behaviour.

67 Machine Learning

Consumption prediction can be formulated as a supervised learning problem.

Let

$$x_i = (r_0, \rho, \kappa, Y, W, \text{Age}, \text{Education}).$$

A machine learning model estimates

$$\hat{c}_i = F(x_i; \theta).$$

67.1 Loss Function

Mean squared error is

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N (c_i - \hat{c}_i)^2.$$

Gradient descent updates parameters according to

$$\theta_{t+1} = \theta_t - \lambda \nabla_{\theta} L.$$

68 Neural Consumption Networks

A neural network representation is

$$h_1 = \sigma(W_1 x + b_1),$$

$$h_2 = \sigma(W_2 h_1 + b_2),$$

$$\hat{c} = W_3 h_2 + b_3.$$

The network learns complex behavioural interactions.

68.1 Interpretation

Neural models can identify:

- Nonlinear credit effects.
- Consumption thresholds.
- Behavioural regime changes.
- Hidden interactions.

69 Reinforcement Learning

Consumers may be modeled as learning agents.

State:

$$s_t = (r_t, \rho_t, \kappa_t, W_t).$$

Action:

$$a_t = c_t.$$

Reward:

$$R_t = U(C_t, S_t).$$

The objective is

$$\max E \left[\sum_{t=0}^{\infty} \gamma^t R_t \right].$$

69.1 Bellman Equation

The value function satisfies

$$V(s) = \max_a [R(s, a) + \gamma E[V(s')]] .$$

Consumption therefore becomes a sequential decision problem.

70 Multi-Agent Consumption Systems

Consider

$$N$$

interacting consumers.

Consumer i influences consumer j through:

- Social pressure.
- Advertising.
- Information.
- Demonstration effects.

- Network contagion.

The consumption function becomes

$$c_i = f\left(r_i, \rho_i, \kappa_i, \sum_j a_{ij}c_j\right).$$

71 Network Consumption Theory

Represent society by a graph

$$G = (V, E).$$

Nodes represent consumers.

Edges represent interactions.

The adjacency matrix is

$$A = (a_{ij}).$$

Consumption propagation follows

$$c_{t+1} = Ac_t.$$

Theorem 21. *Consumption shocks propagate through network structure.*

Proof. Since

$$c_{t+1} = Ac_t,$$

future consumption depends upon adjacency relationships.

Thus shocks spread through the graph.

□

72 Artificial Intelligence and Rationality Credit

Artificial intelligence is a direct producer of rationality credit.

Let

$$AI_t$$

measure AI capability.

Define

$$\rho_T = g(AI_t).$$

Then

$$\frac{\partial \rho_T}{\partial AI_t} > 0.$$

Advanced AI systems increase rationality credit.

72.1 AI-Augmented Consumers

Consumption becomes

$$c = (r_0 + \rho_E + \rho_A + \rho_I + \rho_S + \rho_T)c^* + (1 - r)\eta.$$

AI therefore reduces behavioural deviations.

73 The AI Rationality Theorem

Theorem 22. *If AI systems increase rationality credit, behavioural variance declines.*

Proof. Variance is

$$\text{Var}(c) = \frac{(1 - r)^2}{12}.$$

AI increases

$$r.$$

Therefore

$$\text{Var}(c)$$

falls.

□

74 Consumer Simulation Algorithm

The Complete Theory of Consumption admits direct computational implementation.

74.1 Single Consumer Algorithm

Input:

r0

rho

kappa

alpha

Generate eta in [0,1]

r = min(1,r0+rho)

c = r*alpha + (1-r)*eta

C = c*(1+kappa)*Y

Return:

c,C

75 Population Simulation Algorithm

For each consumer i :

Draw $r0_i$
Draw ρ_i
Draw κ_i
Draw η_i

Compute:

$r_i = \min(1, r0_i + \rho_i)$

$c_i = r_i \cdot \alpha$
 $+ (1 - r_i) \cdot \eta_i$

Compute consumption

Aggregate:

Average consumption
Average savings
Average welfare
Average debt
Average rationality

End

76 Computational Consumption Theory

The Complete Theory of Consumption is particularly suitable for:

- Agent-based modeling.
- Large-scale simulation.
- Machine learning.
- Reinforcement learning.
- Digital twins.
- Economic forecasting.
- Policy analysis.
- AI-assisted governance.

The framework therefore bridges traditional economics and computational social science.

77 General Equilibrium Foundations

The Complete Theory of Consumption may be embedded within a general equilibrium framework.

Consider an economy with:

- Consumers.
- Firms.
- Financial institutions.
- Governments.
- Knowledge-producing institutions.

Consumers maximize utility.

Firms maximize profits.

Financial institutions allocate consumption credit.

Educational and technological institutions generate rationality credit.

77.1 Consumer Optimization

Each consumer solves

$$\max_{C_t, S_t} E \left[\sum_{t=0}^{\infty} \beta^t U(C_t, S_t) \right].$$

subject to

$$W_{t+1} = (1 + i_t)(W_t + Y_t - C_t).$$

and

$$C_t = c_t(1 + \kappa_t)Y_t.$$

77.2 Firm Optimization

Firms solve

$$\max_{K, L} \Pi = F(K, L) - wL - rK.$$

Production is

$$Y = AK^\alpha L^\beta R_N^\theta.$$

Rationality capital therefore enters directly into production.

78 Equilibrium Conditions

A competitive equilibrium satisfies:

1. Utility maximization.
2. Profit maximization.
3. Market clearing.
4. Credit balance.
5. Rationality balance.

Goods market clearing requires

$$Y = C + I + G + NX.$$

Credit market clearing requires

$$\sum_i D_i = \sum_j L_j.$$

Rationality market clearing requires

$$\sum_i \rho_i = R_N.$$

79 The Rationality Market

The Complete Theory introduces an implicit market for rationality.

Producers include:

- Schools.
- Universities.
- Research institutions.
- AI systems.
- Legal systems.
- Statistical agencies.

Consumers acquire rationality credit through participation in these systems.

Definition 17. *The Rationality Market is the collection of institutions that produce, distribute, and maintain rationality credit.*

80 The Rationality Production Function

Let

$$R_N$$

denote national rationality capital.

Suppose

$$R_N = A_R E^\alpha S^\beta T^\gamma I^\delta,$$

where

E = Education,
 S = Science,
 T = Technology,
 I = Institutions.

Theorem 23. *Rationality capital exhibits increasing returns whenever*

$$\alpha + \beta + \gamma + \delta > 1.$$

Proof. The production function is homogeneous of degree

$$\alpha + \beta + \gamma + \delta.$$

If the degree exceeds one, doubling all inputs more than doubles output. □

81 Dynamical Systems Representation

The complete economy may be represented as a dynamical system.

State vector:

$$x_t = (W_t \ \kappa_t \ r_{0,t} \ \rho_t).$$

Dynamics:

$$x_{t+1} = F(x_t).$$

Explicitly,

$$W_{t+1} = (1 + i) [W_t + Y_t - C_t],$$

$$\kappa_{t+1} = (1 + i)\kappa_t + \gamma(c_t - c_t^*),$$

$$r_{0,t+1} = r_{0,t} + \lambda L_t - \mu \delta_t,$$

$$\rho_{t+1} = \rho_t + \psi_t - \chi_t.$$

82 Steady States

A steady state satisfies

$$x_{t+1} = x_t.$$

Therefore

$$W^* = (1 + i) [W^* + Y^* - C^*],$$

$$\kappa^* = (1 + i)\kappa^*,$$

$$r_0^* = r_0^* + \lambda L^* - \mu \delta^*.$$

Theorem 24. *Steady-state intrinsic rationality requires*

$$\lambda L^* = \mu \delta^*.$$

Proof. At equilibrium,

$$r_{0,t+1} = r_{0,t}.$$

Hence

$$\lambda L^* - \mu \delta^* = 0.$$

□

83 Local Stability Analysis

Let

$$x^*$$

be an equilibrium.

Linearizing yields

$$x_{t+1} - x^* J(x_t - x^*),$$

where

$$J = \frac{\partial F}{\partial x}$$

is the Jacobian matrix.

Theorem 25. *The equilibrium is locally stable if every eigenvalue of the Jacobian satisfies*

$$|\lambda_i| < 1.$$

Proof. This follows from standard linear dynamical systems theory.

□

84 Behavioural Phase Diagram

The original framework considered only

$$(r, c).$$

The complete framework introduces

$$(r, \kappa).$$

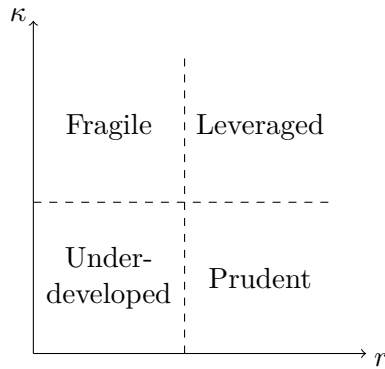


Figure 2: Consumption-credit phase space

85 The Four Regimes Theorem

Theorem 26. *Every economy belongs to one of four principal regimes determined by average rationality and average consumption credit.*

Proof. Partition

$$(r, \kappa)$$

space into quadrants.

Each coordinate is either above or below its threshold.

There are therefore

$$2^2 = 4$$

possible regimes.

□

86 The Consumption Possibility Surface

Consumption is determined by

$$C = c(1 + \kappa)Y.$$

Substituting

$$c = (r_0 + \rho)c^* + (1 - r_0 - \rho)\eta,$$

gives

$$C = \left[(r_0 + \rho)c^* + (1 - r_0 - \rho)\eta \right] (1 + \kappa)Y.$$

This defines a four-dimensional consumption surface.

87 The Fundamental Consumption Equation

Combining all major components yields

$$C_t = \left[(r_{0,t} + \rho_t)c_t^* + (1 - r_{0,t} - \rho_t)\eta_t \right] (1 + \kappa_t)Y_t.$$

This is the central equation of the theory.

Definition 18. *The Fundamental Consumption Equation is*

$$C_t = \left[(r_{0,t} + \rho_t)c_t^* + (1 - r_{0,t} - \rho_t)\eta_t \right] (1 + \kappa_t)Y_t.$$

It combines:

1. Preferences.
2. Rationality.
3. Behavioural impulses.
4. Financial credit.
5. Income.

into a single expression.

88 The Fundamental Consumption Theorem

Theorem 27. *Consumption is jointly determined by:*

1. *Optimal preferences.*
2. *Effective rationality.*
3. *Behavioural disturbances.*
4. *Consumption credit.*
5. *Income.*

Proof. Directly follows from the Fundamental Consumption Equation. □

89 The Rationality-Credit Substitution Theorem

Theorem 28. *Intrinsic rationality and rationality credit are locally substitutable.*

Proof. Consider

$$c = (r_0 + \rho)c^* + (1 - r_0 - \rho)\eta.$$

Differentiating,

$$\frac{\partial c}{\partial r_0} = c^* - \eta,$$

and

$$\frac{\partial c}{\partial \rho} = c^* - \eta.$$

Hence

$$\frac{\partial c}{\partial r_0} = \frac{\partial c}{\partial \rho}.$$

Therefore the two variables are locally substitutable. □

90 The Civilizational Rationality Principle

Theorem 29. *Societies possessing larger stocks of rationality capital achieve superior long-run economic performance, welfare outcomes, and strategic resilience.*

Proof. Rationality capital enters:

1. Consumption efficiency.
2. Production.
3. Wealth accumulation.
4. Welfare.
5. Strategic capability.

Positive marginal contributions in each channel imply superior aggregate outcomes. □

91 A Unified Diagram of Consumption

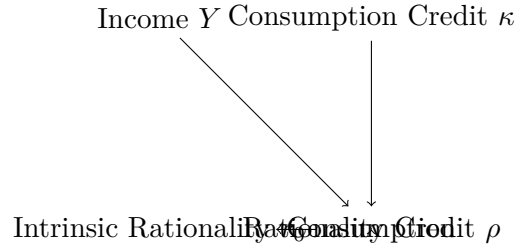


Figure 3: The architecture of the Complete Theory of Consumption

92 Conclusion

This paper developed a unified framework called *The Complete Theory of Consumption*.

The theory extends classical consumption models by introducing two new state variables:

κ = Consumption Credit,

and

ρ = Rationality Credit.

Consumption is determined jointly by:

1. Preferences.
2. Rationality.
3. Behavioural shocks.
4. Financial leverage.
5. Income.

The framework integrates:

- Economics.
- Finance.
- Psychology.
- Philosophy.
- Political science.
- International relations.
- Geopolitics.
- Welfare economics.
- Warfare economics.
- Statistics.

- Computer science.
- Data science.
- Artificial intelligence.

The central contribution is the recognition that modern economies operate simultaneously on financial credit and cognitive credit.

Consumption credit relaxes budget constraints.

Rationality credit relaxes cognitive constraints.

Together they form the foundation of a complete theory of consumption.

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Glossary

r_0 Intrinsic rationality.

ρ Rationality credit.

κ Consumption credit.

δ Rationality debt.

c Propensity to consume.

C Consumption expenditure.
 S Savings.
 Y Income.
 W Wealth.
 R_N National rationality capital.
 R_C Civilizational rationality capital.
 B Behavioural efficiency.
 BEI Behavioural Efficiency Index.
 NX Net exports.
 M_R Military readiness.
 A_G Geopolitical autonomy.
 η Behavioural impulse variable.
 α Consumption preference parameter.
 β Discount factor or regression coefficient depending on context.
 γ Adjustment parameter.
 λ Learning parameter.
 μ Rationality-depreciation parameter.
 AI Artificial intelligence capability.
 SR State rationality production.

The End